ENGINEERNING MATHEMATICS-I [22MA110]

Unit - I

Differential Calculus-I: Polar curves: angle between the radius vector and tangent, angle of intersection of polar curves. Pedal equation for polar curves. Curvature and radius of curvature -Cartesian and pedal forms. (Without proof) **8+3=11 Hours**

Polar Curves

Book work: With usual notations prove that $\tan\phi = r\dfrac{d\theta}{dr}$ **.**

I. Find the angle ϕ between the radius vector and the tangent for each of the following polar curves:

(1) $r^2 \cos 2\theta = a^2$

(2) $r^m = a^m (\cos m\theta + \sin m\theta)$

(3) $r = a(1-\cos\theta)$

(4) $r = a(1+\cos\theta)$ at $\theta = \pi/3$

(5) $(2a/r) = 1-\cos\theta$ at $\theta = 2\pi/3$

(6) $r = a(1+\sin\theta)$ at $\theta = \pi/2$

II. Find the angle of intersection for each of the following pairs of polar curves:

(1) $r = 2\sin\theta$, $r = \sin\theta + \cos\theta$

(2) $r = a(1+\cos\theta)$, $r^2 = a^2 \cos 2\theta$

(3) $r = a(1-cos\theta), r = 2a\, cos\theta$

(4) $r = a\theta, r = a/\theta$

(5) $r^2\, sin\, 2\,\theta = 4,\ r^2 = 16\, sin\, 2\,\theta$

(6) $r = 6\, cos\,\theta\, ,\ r = 2(1 + cos\,\theta)$

III. Show that the following pairs of polar curves intersect orthogonally:

(1) $r = a(1 + cos\,\theta),\ r = b(1 - cos\,\theta)$

(2) $r^n = a^n\, cos\, n\,\theta,\ r^n = b^n\, sin\, n\,\theta$

(3) $r = a\, sec^2(\,\theta/2),\ r = b\, cos\, e\, c^2(\theta/2)$

(4) $r = 2\, sin\,\theta\, ,\ r = 2\, cos\,\theta$

(5) $r^2\, sin\, 2\,\theta = a^2,\ r^2\, cos\, 2\,\theta = b^2$

(6) $r = ae^{\theta},\ \ re^{\theta} = b.$

(7) $r = a(1+\sin\theta)$, $r = b\,(1-\sin\theta)$

IV. Find the pedal equation of the following polar curves:

(1) $r^n = a^n\, cos\, n\,\theta$

(2) $2a/r = (1 + cos\,\theta)$

(3) $r = a(1 + cos\,\theta)$

(4) $r^2 = a^2\, sec\, 2\,\theta$

(5) $r^n = a^n(cos\, n\,\theta + sin\, n\,\theta)$

(6) $r(1 - cos\,\theta) = 2a$

<u>Radius of Curvature in Cartesian Form (without proof)</u>

Find the radius of curvature for the following curves:

(1) $y^2 = \dfrac{4a^2(2a-x)}{x}$ where the curve meets the x-axis.

(2) $y = ax^2 + bx + c$ at $x = \dfrac{1}{2a}\left[\sqrt{a^2-1} - b\right]$

(3) $x^3 + y^3 = 3axy$ at the point $\left(\dfrac{3a}{2}, \dfrac{3a}{2}\right)$

(4) $\sqrt{x}+\sqrt{y}=\sqrt{a}$ at the point where it cuts the line y= x

(5) $x^2 y = a(x^2 + y^{2)}$ at the point (-2a, 2a)

<u>Radius of Curvature in Pedal form (without proof)</u>

1. Show that radius of curvature for the curve $r = a(1+Cos\theta)$ is $\rho = \frac{2}{3}\sqrt{2ar}$ and hence prove that $\frac{\rho^2}{r}$ = constant.
2. Find the radius of curvature for the following:
   (a). $r = a(1 - cos\,\theta)$ (b). $r^n = a^n \, cos\, n\, \theta$
   (c). $r^n = a^n sinn\theta$ (d) $r^2 sin2\theta = a^2$
   (e) $2a/r = 1 + cos\,\theta$